Quick Review: Bits, Types, and Time

- A bit is one binary digit: `0` or `1`.
- C++ types describe how a value is represented and used.
  - Examples: `int`, `float`, `char`, `bool`
  - Type modifiers include `signed`, `unsigned`, `short`, and `long`.
- A hertz (Hz) is one cycle per second.
  - `1 kHz = 1,000 Hz`; `1 MHz = 1,000,000 Hz`.
- Clock rate is one measure of a processor, not a complete speed score.

Measuring Computer Work

Measure	What it counts	Example context
FLOPS	Floating-point operations per second	Scientific calculations
IOPS	Input/output operations per second	Storage devices
Hz	Cycles per second	A clock signal

- FP16, FP32, and FP64 describe floating-point precision formats.
- Floating-point values approximate many decimal fractions; they are useful, not forbidden.
- Division is supported too; its cost depends on the processor and the operation.

Computer Trends

Trend	What it means
More transistors	More components can fit in a chip generation
More cores	A processor can work on multiple instruction streams
More connected	Devices exchange data over networks
More efficient	Designers aim for more work per unit of energy

- These are broad trends, not guarantees for every computer or chip.
- A fast processor is only one part of overall system performance.

Decisions in C++

- A conditional chooses which statements run.
- Use `if` / `else` for conditions and ranges.
- Use `switch` when one value is matched against exact choices.
- Structured branches keep the path through a program readable.

One-Way Decision: `if`

Run the body only when the condition is true.

```
int temperature = 32;

if (temperature > 30) {
    cout << "It is warm.\n";
}
```

- If the condition is false, execution continues after the closing brace.
- A comparison produces `true` or `false`.

Two-Way Decision: `if` / `else`

Exactly one branch runs.

```
int age = 16;

if (age >= 18) {
    cout << "Adult\n";
} else {
    cout << "Not an adult yet\n";
}
```

- The `else` branch handles every case not accepted by the `if` condition.

Several Outcomes: `if` / `else if`

Conditions are tested from top to bottom; the first true branch runs.

```
if (score >= 90) {
  cout << "A\n";
} else if (score >= 75) {
  cout << "B\n";
} else if (score >= 60) {
  cout << "C\n";
} else {
  cout << "Keep practicing\n";
}
```

- Put the most restrictive threshold first.

Relational Operators

Meaning	Operator	Example
Less than	`<`	`3 < 5`
Greater than	`>`	`8 > 2`
Less than or equal	`<=`	`4 <= 4`
Greater than or equal	`>=`	`7 >= 6`
Equal to	`==`	`age == 18`
Not equal to	`!=`	`choice != 0`

- Use `==` to compare; a single `=` assigns a value.

PRACTICE

Lab: Say a Digit

Write a program that reads one integer and prints its English name.

Input	Expected output
`0`	`zero`
`6`	`six`
`9`	`nine`
`-1` or `10`	`Not a single digit`

- Use an `if` / `else if` chain and one final `else` for invalid input.
- Test both ends of the valid range and at least one invalid value.

Exact Choices: `switch`

Use `switch` when one integral value selects among exact cases.

```
switch (day) {
    case 1: cout << "Sunday\n"; break;
    case 2: cout << "Monday\n"; break;
    case 3: cout << "Tuesday\n"; break;
    default: cout << "Invalid day\n";
}
```

- `break` exits the switch; without it, execution can continue into the next case.
- `default` handles values that do not match a listed case.

PRACTICE

Lab: Day of the Week

Read an integer from `1` through `7` and print its day using `switch`.

Input	Expected output
`1`	`Sunday`
`4`	`Wednesday`
`7`	`Saturday`
`0` or `8`	`Invalid day`

- Add a `break` to each day case and a `default` case.
- Check an ordinary case, the last case, and invalid input.

Assignment: Three-Question Quiz

- Write three questions; give each exactly four numbered choices.
- Read the player's choice and report whether it is correct.
- Add one point for each correct answer; the final score is out of 3.
- Reject choices outside `1` through `4` without awarding a point.

Example question: Which city is the capital of the United States?

1. Seattle
2. Los Angeles
3. Washington, D.C.
4. Chicago

- For this first version, ask the three questions one at a time; loops are a later topic.

Challenge: Binary Text to Decimal

- Read a non-empty string containing only `0` and `1`.
- Convert it to a base-10 integer; reject any other character.
- Example: `1011` represents decimal `11`.

Hint: Read digits left to right. For each digit, update the value with `value = value * 2 + digit`.

- This challenge combines strings, loops, and branching.
- Test `0`, `1`, `1011`, and an invalid input such as `10a1`.

Text with `std::string`

- Include `<string>` to use the standard string type.
- A string stores a sequence of characters, such as a name or sentence.
- `cin >> name` reads one word; `getline` can read spaces too.

```
#include <iostream>
#include <string>

int main() {
    std::string name;
    std::cout << "Your name: ";
    std::getline(std::cin, name);
    std::cout << "Hello, " << name << "!\n";
}
```

- If you use `getline` after `cin >>`, handle the leftover newline first.